

C-IMMSIM simulation results

January 5, 2026

Abstract

This document includes the plots relative to the simulation and the outcome of the epitope/peptide prediction used.

Produced by the C-IMMSIM Online server available at
<http://c-immsim.iac.rm.cnr.it> (alias to <http://kraken.iac.rm.cnr.it/C-IMMSIM>)

CITATIONS: For publication of results, please cite the following:

Nicolas Rapin, Ole Lund, Massimo Bernaschi, Filippo Castiglione. Computational Immunology Meets Bioinformatics: The Use of Prediction Tools for Molecular Binding in the Simulation of the Immune System. PLoS ONE 5(4): e9862
doi:10.1371/journal.pone.0009862, 2010

A retrospective validation

In-silico evaluation of adenoviral COVID-19 vaccination protocols: Assessment of immunological memory up to 6 months after the third dose. P. Stolfi, F. Castiglione, E. Mastrostefano, I. Di Biase, S. Di Biase, G. Palmieri, A. Prisco. Frontiers in Immunology, 13 (2022) doi: 10.3389/fimmu.2022.998262
<https://www.frontiersin.org/articles/10.3389/fimmu.2022.998262>

An in vivo validation

Identification and validation of viral antigens sharing sequence and structural homology with tumor associated antigens (TAAs). C. Ragone, C. Manolio, B. Cavalluzzo, A. Petrizzo, A. Mauriello, M-L. Tornesello, F. M. Buonaguro, F. Castiglione, L. Vitagliano, M. Ruvo, M. Tagliamonte, L. Buonaguro. Journal for ImmunoTherapy of Cancer. 9:e002694 (2021)
<https://jitc.bmj.com/content/9/5/e002694>

Original C-IMMSIM model: www.iac.cnr.it/~filippo/c-immsim

GETTING HELP:

Scientific problems: Filippo Castiglione (filippo dot castiglione at cnr dot it)

Technical problems: Ilaria Gonnella (ilaria dot gonnella at cnr dot it)

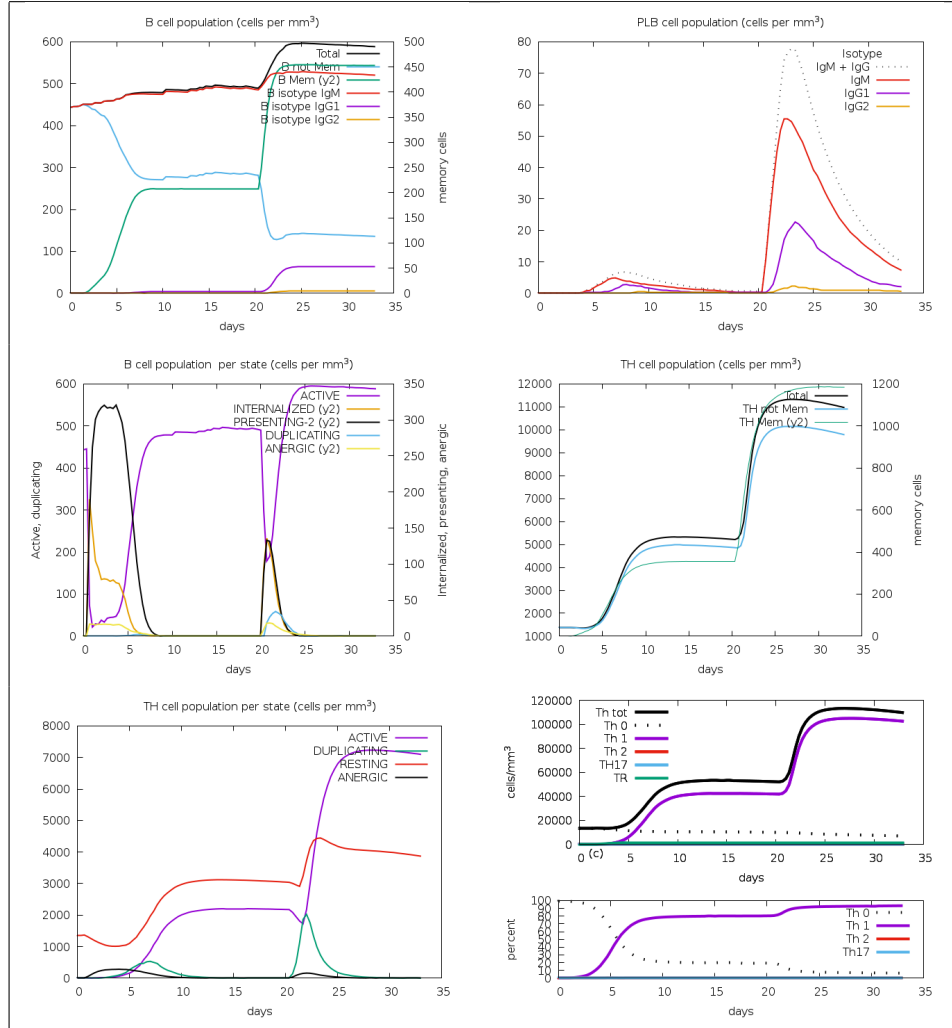


Figure 1: Cell counts shown. Legend: Act=active, Intern=internalized the Ag, Pres II = presenting on MHC II, Dup = in the mitotic cycle, Anergic = anergic, Resting = not active.

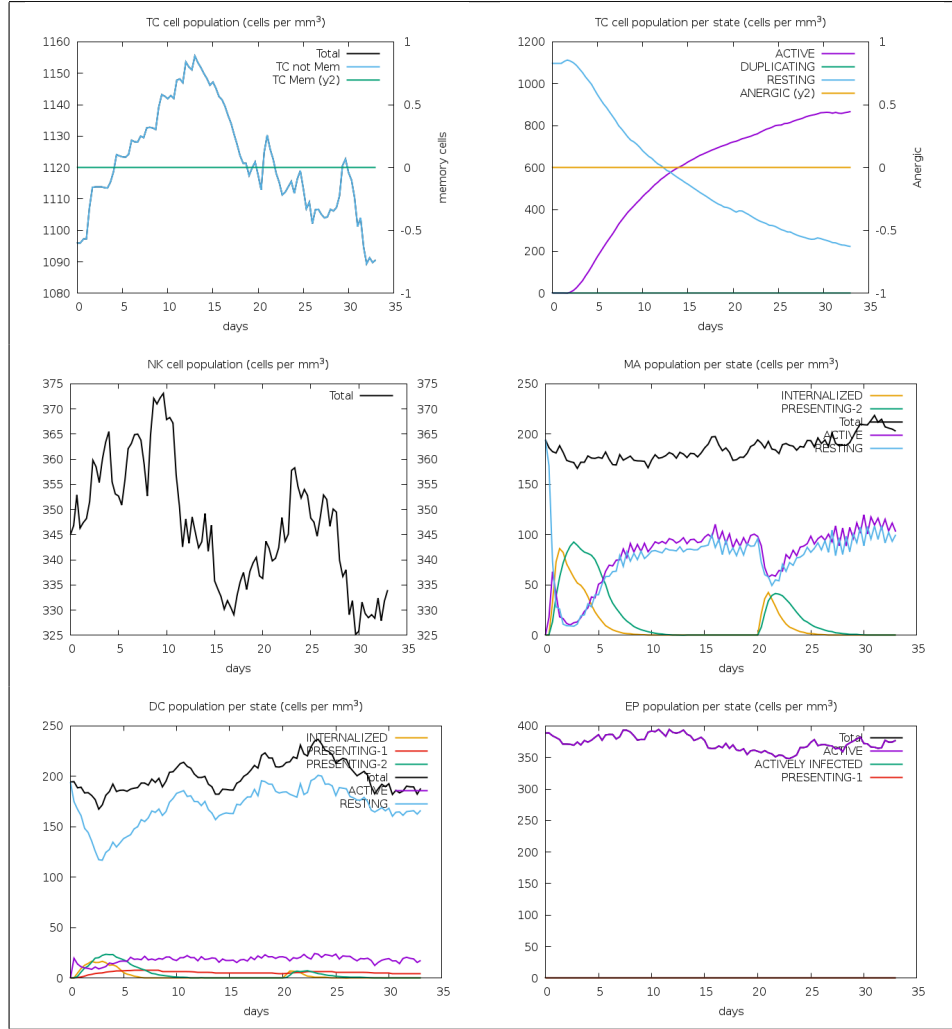


Figure 2: Legend: symbols as figure above.

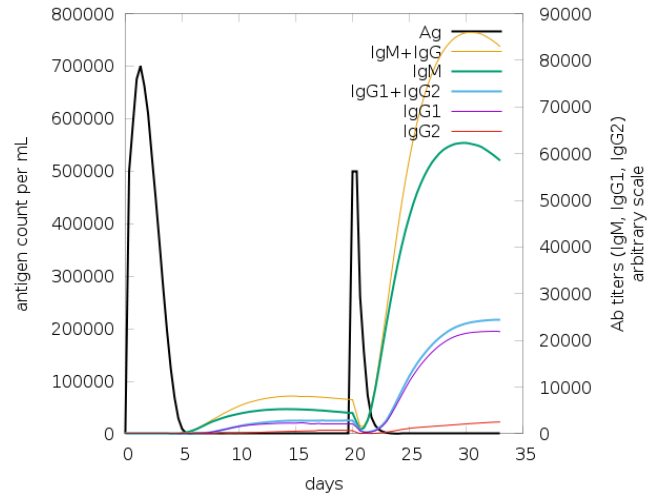


Figure 3: The virus, the immunoglobulins and the immunocomplexes.

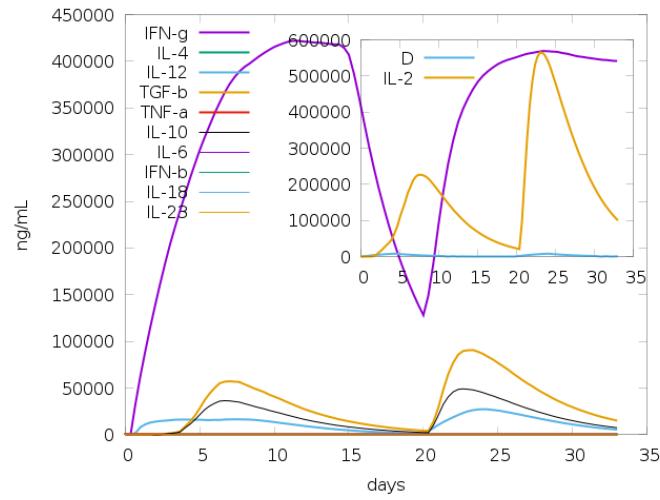


Figure 4: Concentration of cytokines and interleukins. Inset plot shows danger signal together with leukocyte growth factor IL-2.

Use Parker's propensity scale, takes an antigen block as input, and creates a list of residues that are possible epitopes.

APHALSGGSPEPLPQQGLTEAAAKGPGTGPNGLGEKGDTSGEPSGGSGPQRRGGDNHGRGRGRGRGGGRRPGAPGG
SGSGPRHRDGVRRPQKRPSICGCKTHGGTGAGAGAGGAGAGGAGAGGGAGAGGGAGGAGAGGGAGAGGGAGGA
GGAGAGGGAGAEAAAKRVLSVVLLGPGPGTMFEVSVAFGPGPGACMLSAPELEKGPGLPQQGLTAYGPGPGFLDKGTY
TLGPGPGAGTSGWSFAPQAPENAYQAYAAPQLFPVSDITQNQNTNQAGGEAPQPGDNSTVQEAEEKPPM

[illegible]

```

1]      pos=25 len=6      GPGTGP
2]      pos=34 len=53     LGEKGDTSGPEGSGSGPQRRGGDNHGRGRGRGRGRGGRRPGAPGGSGSGPRH
3]      pos=88 len=4      DGVR
4]      pos=104 len=66    GTHGGTGAGAGAGGAGAGGAGAGGGAGAGGGAGGAGGAGGAGAGGGAGAGGGAGGAGAGGGAG
5]      pos=279 len=22    TQNQQTNQAGGEAPQPQDNSTV

```

Given the antigen injected creates the list of peptides for all the NumAgProts proteins and for all i.e., 4 MHC I molecules

Read class I peptide list from file? NO

Allele: A0101
Pseudo sequence: KAVHAEQRNKAQTRA
Threshold: 9.456400
Max score: 29.236000

APPHALSGGSPEPLPQQGLTEAAAKGPGTGPNGLGEKGDTSGPEGSGGSGPQRRGGDNHGRGRGRGRGGGRRPGAPGG
SGSGPRHRDGVRRPQKRPSICGCKGTHGGTGAGAGAGGAGAGGAGAGGGAGAGGGAGGAGGAGAGGGAGAGGGAGGA
GGAGAGGGAGAEAAAKRVLSVVLLGPGPGTMFEVSVAFGPGPGACMLSAPELEKGPGLPQQGLTAYGPGPGFLDKGTY
TLGPGPGAGTGSWFSAPQAPENAYQAYAAPQLFPVSDITQNOQTNQAGGEAPQPGDNSTVQEAALKPPM

```
0] pos= -1 score=1.000000 unnormalised=104.8410000000 non-binding event
```

Allele: A0101
Pseudo sequence: KAVHAEQRNKAQTRA
Threshold: 9.456400
Max score: 29.236000

Antigen sequence file: /opt/lampp/htdocs/C-IMMSIM/Jobs/input/44265_20260105-015520_5_KsDp84vhkdyd.FSA_1_001

APPHALSGGSPEPLPQQQLTEAAAKGPGTGPNGLGEKGDTSGPEGSGGSGPQRRGGDNHGRGRGRGRGRGGGRPGAPGG
SGSGPRHRDGVRPQKRPSICIGKGTGAGAGAGGAGAGGAGAGGAGAGGAGAGGAGAGGAGAGGAGAGGAGAGGAGGAGGA
GGAGAGGGAGAAAKRVLSVVLLGPGPGTMFEVSVAFGPGPGACMLSAPLEKGPGLPQQQLTAYGPGPGFLDKGTY
TLGPGPGAPTGSWFSAPQAPENAYQAYAAPQLFPVSDITQNNQTNQAGGEAPQPGDNSTVQEAANKPPM

Epitopes of protein 0 -----

0] pos= -1 score=1.000000 unnormalised=104.8410000000 non-binding event

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Allele: B0702

Pseudo sequence: KAAREEQIKAQTRE

Threshold: 8.702800

Max score: 28.406000

Antigen sequence file: /opt/lampp/htdocs/C-IMMSIM/Jobs/input/44265_20260105-015520_5_KsDp84vhkdyd.FSA_1_001

APPHALSGGSPEPLPQQQLTEAAAKGPGTGPNGLGEKGDTSGPEGSGGSGPQRRGGDNHGRGRGRGRGRGGGRPGAPGG
SGSGPRHRDGVRPQKRPSICIGKGTGAGAGAGGAGAGGAGAGGAGAGGAGAGGAGAGGAGAGGAGAGGAGAGGAGGA
GGAGAGGGAGAAAKRVLSVVLLGPGPGTMFEVSVAFGPGPGACMLSAPLEKGPGLPQQQLTAYGPGPGFLDKGTY
TLGPGPGAPTGSWFSAPQAPENAYQAYAAPQLFPVSDITQNNQTNQAGGEAPQPGDNSTVQEAANKPPM

Epitopes of protein 0 -----

0] pos= 13 score=0.005862 unnormalised=0.6622000000 LPQGGQLTEA
1] pos= 73 score=0.002667 unnormalised=0.3012000000 RPGAPGGSG
2] pos= 92 score=0.017106 unnormalised=1.9322000000 RPQKRPSCI
3] pos= 216 score=0.013875 unnormalised=1.5672000000 GPGLPQQQL
4] pos= 247 score=0.021019 unnormalised=2.3742000000 APTGSWFSA
5] pos= 255 score=0.002649 unnormalised=0.2992000000 APQPAPENA
6] pos= 291 score=0.008660 unnormalised=0.9782000000 APQPGDNST
7] pos= -1 score=0.928163 unnormalised=104.8410000000 non-binding event

=====

Allele: B0702

Pseudo sequence: KAAREEQIKAQTRE

Threshold: 8.702800

Max score: 28.406000

Antigen sequence file: /opt/lampp/htdocs/C-IMMSIM/Jobs/input/44265_20260105-015520_5_KsDp84vhkdyd.FSA_1_001

APPHALSGGSPEPLPQQQLTEAAAKGPGTGPNGLGEKGDTSGPEGSGGSGPQRRGGDNHGRGRGRGRGRGGGRPGAPGG
SGSGPRHRDGVRPQKRPSICIGKGTGAGAGAGGAGAGGAGAGGAGAGGAGAGGAGAGGAGAGGAGAGGAGAGGAGGA
GGAGAGGGAGAAAKRVLSVVLLGPGPGTMFEVSVAFGPGPGACMLSAPLEKGPGLPQQQLTAYGPGPGFLDKGTY
TLGPGPGAPTGSWFSAPQAPENAYQAYAAPQLFPVSDITQNNQTNQAGGEAPQPGDNSTVQEAANKPPM

Epitopes of protein 0 -----

0] pos= 13 score=0.005862 unnormalised=0.6622000000 LPQGGQLTEA
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5] pos= 255 score=0.002649 unnormalised=0.2992000000 APQPAPENA

6]	pos= 291	score=0.008660	unnormalised=0.9782000000	APQPGDNST
7]	pos= -1	score=0.928163	unnormalised=104.8410000000	non-binding event

DoPeptideList_II:

Given the antigen injected creates the list of peptides for all the
NumAgProts proteins and for all i.e., 2 MHCII molecules

Read class II peptide list from file? NO

Allele: DRB1_0101
Pseudo sequence: KAFHVEQRKAQTRV
Threshold: 2.392440
Max score: 26.461000

Antigen sequence file: /opt/lampp/htdocs/C-IMMSIM/Jobs/input/44265_20260105-015520_5_KsDp84vhkdyd.FSA_1_001

APPHALSGGSPEPLPQQQLTEAAAKGPGTGPNGLGEKGDTSGPEGSGGSGPQRRGGDNHGRGRGRGRGRGGGRPGAPGG
SGSGPRHRDGVRRPQKRPSICGCKGTHGGTGAGAGAGGAGAGAGGGAGAGGAGGAGAGGAGAGGAGAGGGAGAGGGAGGA
GGAGAGGGAGAEAAAKRVLSVVLLGPGPGTMFEVSVAFGPGPGACMLSAPLEKGPGLPQQQLTAYGPGPGFLDKGTY
TLGPGPGAPTGSWFSAPQAPENAYQAYAAPQLFPVSDITQNNQTNQAGGEAPQPGDNSTVQEAAAKPPM

Epitopes of protein 0 -----

0]	pos= 5	score=0.035276	unnormalised=3.1995600000	LSGGSPEPL
1]	pos= 181	score=0.018452	unnormalised=1.6735600000	VVLLGPGPG
2]	pos= 182	score=0.009389	unnormalised=0.8515600000	VLLGPGPGT
3]	pos= 198	score=0.032300	unnormalised=2.9295600000	FGPGPGACM
4]	pos= 211	score=0.009830	unnormalised=0.8915600000	LEKGPGL
5]	pos= 239	score=0.069191	unnormalised=6.2755600000	YTLGPGPGA
6]	pos= 241	score=0.002608	unnormalised=0.2365600000	LGPGPGAPT
7]	pos= 264	score=0.093425	unnormalised=8.4735600000	YQAYAAPQL
8]	pos= -1	score=0.729530	unnormalised=66.1680000000	non-binding event

Allele: DRB1_0101
Pseudo sequence: KAFHVEQRKAQTRV
Threshold: 2.392440
Max score: 26.461000

Antigen sequence file: /opt/lampp/htdocs/C-IMMSIM/Jobs/input/44265_20260105-015520_5_KsDp84vhkdyd.FSA_1_001

APPHALSGGSPEPLPQQQLTEAAAKGPGTGPNGLGEKGDTSGPEGSGGSGPQRRGGDNHGRGRGRGRGRGGGRPGAPGG
SGSGPRHRDGVRRPQKRPSICGCKGTHGGTGAGAGAGGAGAGAGGGAGAGGAGGAGAGGAGAGGAGAGGGAGAGGGAGGA
GGAGAGGGAGAEAAAKRVLSVVLLGPGPGTMFEVSVAFGPGPGACMLSAPLEKGPGLPQQQLTAYGPGPGFLDKGTY
TLGPGPGAPTGSWFSAPQAPENAYQAYAAPQLFPVSDITQNNQTNQAGGEAPQPGDNSTVQEAAAKPPM

Epitopes of protein 0 -----

0]	pos= 5	score=0.035276	unnormalised=3.1995600000	LSGGSPEPL
1]	pos= 181	score=0.018452	unnormalised=1.6735600000	VVLLGPGPG
2]	pos= 182	score=0.009389	unnormalised=0.8515600000	VLLGPGPGT
3]	pos= 198	score=0.032300	unnormalised=2.9295600000	FGPGPGACM
4]	pos= 211	score=0.009830	unnormalised=0.8915600000	LEKGPGL

5]	pos= 239	score=0.069191	unnormalised=6.2755600000	YTLGPGPGA
6]	pos= 241	score=0.002608	unnormalised=0.2365600000	LGPGPGAPT
7]	pos= 264	score=0.093425	unnormalised=8.4735600000	YQAYAAPQL
8]	pos= -1	score=0.729530	unnormalised=66.1680000000	non-binding event
